MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, HIGH-SPEED
FLIP-FLOPS, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-0038510/22C(USAF), dated 1 December 1975.

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TABLE I, device type 02: Delete and substitute new table as printed on page 2 of this amendment.

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TABLE I, device type 03: Delete and substitute new table as printed on page 3 of this amendment.

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(Project 5962-F088-62)

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TABLE I. Electrical characteristics tests (device type 02).

Test No. 1/ 2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	Input voltage required to ensure a high level input (each input)	V_{IH}	V_{CC} = 4.5 V	2			V
2	Input voltage required to ensure a low level input (each input)	V_{IL}	V_{CC} = 4.5 V		0.8		V
3*	High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2 V, I_{OH} = -500 µA	2.4		±.12	V
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2 V, I_{OL} = 20 mA		0.4	±.04	V
5	Low level input current at J, K, or clock	I_{IL1}	V_{CC} = 5.5 V	-1	-2		mA
6	Low level input current at clear	I_{IL2}	V_{IN} = 0.4 V	-2	-4		mA
7*	High level input current at J, K, or clock	I_{IH1}	V_{CC} = 5.5 V		50	±5.0	µA
8*	High level input current at clear	I_{IH2}	V_{IN} = 2.4 V		100	±10.0	µA
9	High level input current at J, K, or clock	I_{IH3}	V_{CC} = 5.5 V		100		µA
10	High level input current at clear	I_{IH4}	V_{IN} = 5.5 V		200		µA
11	Short circuit output current	I_{OS}	V_{CC} = 5.5 V	-40	-100		mA
12	Supply current	I_{CC}	V_{CC} = 5.5 V		50		mA
13	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -8 mA T_A = 25°C		-1.5		V

1/ Delta computations shall be performed on parameters marked with an asterisk (*) with data read and recorded.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE I. Electrical characteristics tests (device type 03).

Test No.	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ)	Units
				Min	Max		
1	Input voltage required to ensure a high level input (each input)	V_{IH}	V_{CC} = 4.5 V	2			V
2	Input voltage required to ensure a low level input (each input)	V_{IL}	V_{CC} = 4.5 V		0.8		V
3*	High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2 V, I_{OH} = -1 mA	2.4		±.12	V
4*	Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.8 V, V_{IH} = 2 V, I_{OL} = 20 mA		0.4	.04	V
5	Low level input current at preset or D	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	-1	-2		mA
6	Low level input current at clear or clock	I_{IL2}		-2	-4		mA
7*	High level input current at D	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V		50	±5	µA
8*	High level input current at preset or clock	I_{IH2}			100	±10	µA
9*	High level input current at clear	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		150	±15	µA
10	High level input current at D	I_{IH4}			100		µA
11	High level input current at preset or clock	I_{IH5}			200		µA
12	High level input current at clear	I_{IH6}			300		µA
13	Short circuit output current	I_{OS}	V_{CC} = 5.5 V	-40	-100		mA
14	Supply current	I_{CC}	V_{CC} = 5.5 V		50		mA
15	Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -8 mA, T_A = 25°C		-1.5		V

1/ Delta computations shall be performed on parameters marked with an asterisk (*) with data read and recorded.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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